

CS614 Final Term Solved Current paper 2023 by Masters

Data Warehousing (CS614)

Question: **1** (Marks: 1)

Attempted Questions: **0** Total Questions: **50**

The degree of similarity between two records, often measured by a numerical value between _____, usually depends on application characteristics.

Choices:

☒ 0 and 1



☐ 0 and 10

☐ 0 and 100

☐ 0 and 99

Masters

Data Warehousing (CS614)

Question: **2** (Marks: 1)

Attempted Questions: **1** Total Questions: **50**

The _____ measures the ratio of desired outcomes to total outcomes.

Choices:

☒ Simple Ratio



Masters

☐ Min Operation

☐ Max Operation

☐ Weighted Average



Data dependencies between different phases of computation introduce synchronization requirements that force _____ execution.

Choices:

☒ Sequential

✓ *Masters*

☐ Parallel

☐ Analogous

☐ Interactive Parallel

Which is the least appropriate join operation for Pipeline parallelism?



Choices:

Hash Join

☐

Inner Join

☐

Outer Join

☐

Sort-Merge Join



Masters

Classification consists of examining the properties of a newly presented observation and assigning it to a predefined _____.

Choices:

Class

☐

Masters

Object

☐

Container

☐

Subject

☐

Majority of data warehouse projects fail due to the complexity of the

_____.



Choices:

☐ Development Process

✓ *Masters*

☐ Analytical Process of Cube

☐ Query Complexity

☐ Indexing Complexity

The application development quality-assurance activities cannot be completed until the data is _____.

Choices:

☒ Stabilized



☐ Identified

☐ Finalized

☐ Computerized

In the context of Business Development Lifecycle (Kimball's approach), the first task in technology track is _____

Choices:

☒ Technical Architecture Design

☐ Requirement Specification Development

☐ Requirement Analysis

☐ Lifecycle Model Selection

Focusing on data warehouse delivery only, often ends up _____.

✓

Choices:

Rebuilding

☒

Success

☐

Good Stable Product

☐

None of the given options

☐

Execution of a package can be completed successfully or it may be stopped due to some error.
In case of successful completion of execution all the transactions will be _____.

Choices:

☐ Committed to the database ✓

☒ Rolled back ✗

☐ Check marked

☐ Stubbed

Identify the TRUE statement about Hypertext Transfer Protocol (HTTP).



Choices:

☒ HTTP is stateless protocol



☐ HTTP is not a word wide web protocol

☐ HTTP is used to maintain sessions

☐ HTTP is message routing protocol

The i th bit is set to 1 if the i th row of the base table has the value for the indexed column.
This statement refers to:



Choices:

Inverted index

☐

Bitmap index

☒

Dense index

☐

Sparse index

☐

Which of the following is NOT one of the nested loop join techniques?

Choices:

Naïve nested-loop join

☐

Index nested-loop join

☐

Block nested-loop join

☒

Temporary index nested-loop join

☐

Which of the following is NOT one of the data mining techniques?



Choices:

Clustering

☐

Micro filtering

☒

Decision Trees

☐

Rule induction

☐

For a K-means clustering algorithm:

Choices:

☐ t is much smaller than n



☐ t is greater than n

☐ t much greater than n

☐ t is equal to square of n

In DWH projects, it is assured that _____ environment is similar to the production environment.

✓

Choices:

☐ Planning

☒ Development



☐ Analysis

☐ Requirement extraction

In the Information Age, the _____ learning organization is at a distinct disadvantage. This term means "impaired functioning".

Choices:

Functional

☐

Dysfunctional

☒

Purposeful

☐

Serviceable

☐

Major issues of data cleansing had arisen due to data processing and handling at _____ levels by different groups of people.



Choices:

Two

☐

Four

☒

Five

☐

Six

☐

_____ is a small piece of information generated by the Web server and stored on the client machine.

Choices:

Session

☐

Log file

☐

Cookie

☒

Timestamp

☐

_____ can be placed in front of our enterprise's Web servers to help them offload requests for frequently accessed content.

Choices:

Browser caches

☐

Reverse Proxy

☒

Log file

☐

Forward Proxy

☐

In the process of data extraction, the wizard creates package which is a collection of



Choices:

Data

☐

SQL Commands

☐

Reports

☐

Tasks



As soon as transformation tab is clicked, designer tries to map the source and destination table _____ automatically on the base of resemblances in the names.

Choices:

☒ Columns



☐ Rows

☐ Tables

☐ Data

To identify the _____ required, we need to perform data profiling.

Choices:

Complexity

☐

Cost

☐

Time

☐

Degree of Transformation

☐

If some error occurs, execution of transactions will be terminated abnormally and all transactions will be _____.

Choices:

☐ Committed to the database

☐ Check marked

☐ None of the given options

☒ Rolled back



Semantically "Dirty Data" class of anomalies includes:

- I) Lexical Errors
- II) Integrity Constraints Violation
- III) Business Rule Contradiction
- IV) Irregularities

Choices:

☐ (I) and (II) only

☐ (I), (II), and (III)

☐ (II), (III), and (V) only

☒ (I), (IV), and ~~(V)~~ only

PTCL is an example of _____ data warehouse organization.



Choices:

Government

☐

Financial Service

☐

Autonomous

☐

Telecommunication

☐

Pakistan is one of the five major _____ country in the world.



Choices:

☒ Cotton-growing



☐ Rice-growing

☐ Weapon Producing

☐ Apple-growing

In agriculture extension, _____ is that pest population beyond which the benefit of spraying outweighs its cost.

Choices:

☐ Medicine Threshold Level

☒ Economic Threshold Level

☐ Profit Threshold Level

☐ Loss Threshold Level

Data recorded by pest scouts consists of two parts:



Choices:

☐ Valid and Invalid

☒ Static and Dynamic

☐ Volatile and Non-Volatile

☐ Clean and dirty

Which of the following is/are drawback(s) of traditional web searches?

Choices:

- ☒ Limited to keyword based matching 
- ☐ Can not distinguish between the contexts in which a link is used
- ☐ Coupling of files has to be done manually 
- ☐ Do not perform searches quickly

Which of the following is NOT an issue of Clickstream data?



Choices:

☐ Identifying the Visitor Origin

☐ Identifying the Session

☐ Identifying the Visitor

☒ Identifying the domain server



Data Transformation Services (DTS) allows us to connect through any data source or destination that is supported by _____.



Choices:

☒ OLE DB



☐ OLTP

☐ OLAP

☐ Data warehouse

In the context of web warehousing, which of the following is NOT a way to identify the session?

Choices:

☐ Using Time-contiguous Log Entries

☐ Using Transient Cookies

☐ Using HTTP's secure sockets layer (SSL)

☒ Using Asynchronous session track (AST) protocol

_____ is a process, which involves gathering of information about column through execution of certain queries with intention to identify erroneous records.



Choices:

☒ Data Profiling



☐ Data Anomaly Detection

☐ Record Duplicate Detection

☐ Data Cleaning

In the context of Data standardization, data profiling is performed _____, in order to judge the effectiveness of transformation of data elements.

Choices:

Once

☐

Two times

☒

Three times

☐

Four times

☐

_____ improve(s) the functional processes used to create, manage, access, and use data.

Choices:

☒ Process Improvement



☐ System Improvement

☐ Policy & Procedure Improvement

☐ Data Design Improvement

If a dense index fits into memory, then it costs only _____ disk I/O access to locate a record by the given key.

Choices:

☒ One☐ Two☐ $\log(n)$ ☐ n

Kimball's Approach for data warehouse implementation is _____.



Choices:

☐ Data-Driven

☒ Goal-Driven

☐ User-Driven

☐ Method-Driven

The product Selection phase falls _____ of the Business Dimensional Lifecycle (Kimball's Approach).



Choices:

☐ Lifecycle Technology Track



☐ Lifecycle Data Track

☐ Lifecycle Analytic Applications Track

☐ Lifecycle development track

NADRA is an example of _____ data warehouse organization.



Choices:

☐ Telecommunication

☐ Educational

☐ Transportation

☒ Government



Data Warehousing (CS614)

Question: 41 (Marks: 3)

Attempted Questions: 40 Total Qu

There are different metrics used for comparison of data mining techniques e.g. accuracy, speed, robustness, scalability and interpretability. Given below are two statements, each of which corresponds to some metric. Name the metric, each statement corresponds to.

- a) Ability of the technique to work accurately even in conditions of noisy or dirty data.
- b) Measure of correctness of your model.

Answer:



- a) Robustness
- b) Accuracy

Data Warehousing (CS614)

Question: **42** (Marks: 3)

Attempted Questions: **41** Total Questions: **50**

Specify at least three dimensions common in context of Web data warehouse.

Answer:



|



Specify at least three dimensions common in context of Web data warehouse.

1. Date
2. Time of day
3. Part Vendor
4. Transaction
5. Status

Answer:



1. Time dimension
2. User dimension
3. Geographical dimension

I

Data Warehousing (CS614)

Question: 43 (Marks: 3)

Attempted Questions: 42 Total Qu

Write down the names of any two Implementation strategies for data warehouse project.

Answer:



Top-Down Implementation

Bottom-up Implementation

Data Warehousing (CS614)

Question: **44** (Marks: 3)

Attempted Questions: **43** Total Questions: **50**

Incorrect records should be corrected before the transformation process; we can update the records through SQL Queries or through manual checking.

Identify which technique (SQL Queries or manual checking) is best suitable for each of the following cases:

1. Total number of records is 5
2. Total number of records is 5000

Answer:



1. Manually Checking because record is small and easy to check.

2. SQL Queries because there is large amount of data that will be difficult to check manually.

Data Warehousing (CS614)

Question: 45 (Marks: 3)

Attempted Questions: 44 Total Questions: 50

From Dense and Sparse Indexes, identify the correct index for each of the following:

1. If fits in the memory, costs only one disk I/O access to locate a record given a key.
2. Uses less space, but the block has to be searched, even for unsuccessful searches.



Answer:



1. Dense Index

2. Sparse Index

List down the first two steps of the **Basic Sorted Neighborhood (BSN)** Method?



Answer:



Steps 1: Create Keys

Compute a key for each record in the list by extracting relevant fields or portions of fields

Effectiveness of the this method highly depends on a properly chosen key

Step 2: Sort Data

Sort the records in the data list using the key of step 1

Step 3: Merge

Move a fixed size window through the sequential list of records limiting the comparisons for matching records to those records in the window

Question: **46** (Marks: 3)

List down the first two steps of the **Basic Sorted Neighborhood (BSN)** Method?

Answer:



Following are two steps:

1. Data ordering
2. Divide into neighborhoods



Data Warehousing (CS614)

Question: **47** (Marks: 5)

Attempted Questions: **46** Total Questions: **50**

List down only names of the four techniques used for "Data Quality Validation".

Answer:



Data Quality Validation Techniques

1. Referential Integrity (RI).
2. Attribute domain.
3. Using Data Quality Rules.
4. Data Histograming.

exponent can be included.

22.2 Data Quality Validation Techniques

- Referential Integrity (RI).
- Attribute domain.
- Using Data Quality Rules.
- Data Histograming.

Data Warehousing (CS614)

Question: **48** (Marks: 5)Attempted Questions: **47** Total Questions: **50**

Consider the following (test) data table:

Name	Time	Item	Gender
Younas	19	2	Male
Yasin	11	3	Male
Waqas	3	1	Male

Now suppose that from our training data, we predicted the following classification model to predict the unknown gender of a person.

If $(\text{Time} / \text{Item}) > 6$

Gender = "Female"

Else

Gender = "Male"

Using the above test data and estimation model:

- Calculate the % accuracy of the given model (Suppose the test data comprises of only three records given above.)
- For a person named "Ali", Time = 14, Item = 2, calculate the gender according to the given model.

Answer:



a) **Calculate the % accuracy of the given model:**

- For Younas: $(19/1)^2 > 6 \Rightarrow$ Predicted Gender: Female (Incorrect)
- For Yasin: $(11/3) \leq 6 \Rightarrow$ Predicted Gender: Male (Correct)
- For Waqas: $(1/1)^3 \leq 6 \Rightarrow$ Predicted Gender: Male (Correct)

Accuracy = (Correct Predictions / Total Predictions) * 100

Accuracy = $(2 / 3) * 100 = 66.67\%$

b) **For a person named "Ali", Time = 14, Item = 2:**

- $(14/2) > 6 \Rightarrow$ Predicted Gender: Female



Data Warehousing (CS614)

Question: **49** (Marks: 5)

Attempted Questions:

Name any five data warehouse Development methodologies.

Answer:



Development methodologies

- Waterfall model
- Spiral model
- RAD Model
- Structured Methodology
- Data Driven
- Goal Driven
- User Driven

Following is the header (Column names with descriptions) of student table stored in SQL Server:

Reg#: Student identity

Name: Student name

Father: Father name

Address: Permanent address

Date of Birth: Date of Birth

lastDeg: Last degree achieved

Reg Date: Date of Enrollment

Reg Status: Status of Enrollment (A/T)

Degree_Status: Status of Degree (C/I)

Dirty: Correctness of record 1 means incorrect and 0 is correct (0/1)

You are required to write the SQL queries to perform the following tasks:

- Count all the records for which we should validate business rules
- Count all Incorrect records in the table

Answer:



1. Count all the records for which business rules should be validated:

sql

 Copy code

```
SELECT COUNT(*) AS TotalRecordsToValidate
FROM student
WHERE Reg Status = 'A' AND Degree Status = 'C' AND Dirty = 0;
```

1. Count all Incorrect records in the table:

sql

 Copy code

```
SELECT COUNT(*) AS IncorrectRecords  
FROM student  
WHERE Dirty = 1;
```